# County-Level MSM Population Estimation

We estimate the population of men who have sex with men (MSM) at the county level using a multi-step approach that incorporates household structure, urbanicity stratification, and demographic weighting. The method builds on prior work by Grey et al. and Oster et al., with a slight adjustment in methods and sources and extending the same algorithm for prediction of demographic subgroup populations by Age also to Income, Race, and Education.

## 1. Estimation of Same-Sex Male Households

We assume that the ratio of male-male (MM) to total same-sex (MM + FF) households is consistent across counties within a given urbanicity and state. To operationalize this, we compute the following proportion for each state $s$ and urbanicity $u$:

$$\rho_{s,u} = \frac{\sum_{c \in s,u} \mathrm{MM}_c}{\sum_{c \in s,u} (\mathrm{MM}_c + \mathrm{FF}_c)} \tag{1}$$

where:

- $\mathrm{MM}_c$: Number of male-male same-sex households in county $c$
- $\mathrm{FF}_c$: Number of female-female same-sex households in county $c$

We then use this value to estimate the number of same-sex male (SSM) households in each county as:

$$\widehat{\mathrm{SSM}}_c = \rho_{s,u} \cdot \mathrm{SS}_c \tag{2}$$

where $\mathrm{SS}_c$ is the total number of same-sex households in county $c$.

**Assumption:** This adjustment assumes that the MM/(MM+FF) composition is stable within each state-urbanicity pairing and reflects meaningful differences in couple distributions by region.

**Imputation for Missing SSM Household Counts** In cases where $\widehat{\mathrm{SSM}}_c$ is missing due to zero-suppression or data unavailability, we impute values using urbanicity-stratified averages. Specifically, we:

Add an imputed value based on the average SSM household rate for the urbanicity stratum:

$$\widehat{\mathrm{SSM}}_c \leftarrow \widehat{\mathrm{SSM}}_c + \left( \frac{\sum_{c \in u} \widehat{\mathrm{SSM}}_c}{\sum_{c \in u} \mathrm{HH}_c} \right) \cdot \mathrm{HH}_c$$

where $u$ denotes the urbanicity group and $\mathrm{HH}_c$ is the total number of households in county $c$.

This ensures consistency in household-based MSM estimation across counties with suppressed or incomplete SSM data.

## 2. MSM Household Index

We construct a household-based index to scale county-level estimates relative to urbanicity-wide norms:

$$\text{Index}_c = \frac{\widehat{\text{SSM}}_c/\text{HH}_c}{\overline{\text{SSM}}_u/\overline{\text{HH}}_u} \tag{3}$$

where:

- $\text{HH}_c$: Total households in county $c$
- $\overline{\text{SSM}}_u$, $\overline{\text{HH}}_u$: Aggregated SSM households and total households in urbanicity $u$

## 3. Base MSM Proportion by Urbanicity

From the General Social Survey (GSS), we extract urbanicity-stratified base MSM proportions:

$$P_u = \mathbb{P}(\text{MSM} \mid u) \tag{4}$$

## 4. Initial County-Level MSM Estimate

We estimate the MSM population in county $c$ as:

$$\widehat{M}_c = P_u \cdot \text{MalePop}_c \cdot \text{Index}_c \tag{5}$$

where $\text{MalePop}_c$ is the number of adult males in county $c$, taken from the ACS.

## 5. Demographic Breakdown

To distribute county-level MSM estimates into demographic subgroups (e.g., by age, education, income), we use urbanicity-specific subgroup proportions from the GSS:

1. Compute unadjusted subgroup estimates:

$$\widehat{M}_{c,d} = p_{u,d} \cdot \text{MalePop}_{c,d}$$

2. Normalize subgroup shares within county:

$$s_{c,d} = \frac{\widehat{M}_{c,d}}{\sum_{d'} \widehat{M}_{c,d'}}$$

3. Scale to match county total MSM:

$$\widehat{M}_{c,d} = s_{c,d} \cdot \widehat{M}_c$$

At both the county and demographic subgroup levels, we calculate the proportion of MSM relative to the total adult male population.

## 6. Simulation Method

We simulate 100,000 draws using the appropriate distributions (Table 1) and then pull our summary values: mean, median, and the 95 percent confidence interval.

| **Variable Simulated** | **Distribution Used** |
|---|---|
| Male population (county-level) | Normal (mean from ACS, SE from MOE/1.645) |
| Male population (age, education, income groups) | Normal (group-specific estimates, with SE from MOE) |
| Total households (county-level) | Normal (with SE from MOE) |
| Same-sex households (MM, FF, SS) | Log-normal (after log-transform of mean and SE) |
| MSM proportion (by urbanicity) | Logit-normal |
| MSM proportion (by demo × urbanicity) | Logit-normal |
| MM / (MM + FF) ratio | Logit-normal (logit transformed ratio, SE derived from counts) |

**Table 1:** Simulated input quantities and assumed distributions for uncertainty propagation

We use the following transformation to simulate draws from a logit-normal distribution:

$$\mu = \text{logit}(p), \quad \text{SE}_{\text{logit}} = \frac{\text{SE}}{p(1-p)}, \quad p^{(\text{draw})} = \text{logit}^{-1}\left(\mathcal{N}(\mu, \text{SE}_{\text{logit}})\right)$$

Any simulated values that resulted in negative estimates (e.g., due to draws from the normal distribution) were set to zero to preserve interpretability and validity.

## 7. Data Sources

The data sources are detailed below.

| Data Source | Variable(s) | Usage in MSM Estimation |
|---|---|---|
| **ACS 2022** | Total households ($HH_c$)<br>Male population 18+ ($MalePop_c$) | Denominator for SSM index<br>Base population for MSM count calculation |
| **ACS 2023 (Age, Educ), 2021 (Income)** | Demographic distributions: age, education, income | Weighting male population by subgroup for demographic breakdown |
| **ACS 2014** | Same-sex male households ($MM_c$),<br>Female same-sex ($FF_c$),<br>Total same-sex ($SS_c$) | Used to compute MM/SS ratio<br>Derivation of SSM index |
| **ACS 2022 (for SS)** | Same-sex households (SS) | Used to update total SS counts by county |
| **GSS 2016–2024** | MSM prevalence by urbanicity ($P_u$)<br>MSM prevalence by subgroup and urbanicity ($p_{u,d}$) | Determines baseline and demographic-specific MSM rates |
| **NCHS Urban-Rural Classification** | Urbanicity designation ($u$) for each county | Stratifies counties for applying appropriate MSM proportions |

**Table 2:** Data sources, variables, and their roles in MSM population estimation

# Details on Data Cleaning, Harmonization, and Assumptions

Prior to estimation, we applied several steps to clean and harmonize the demographic and behavioral data used in the MSM estimation pipeline.

## GSS Data Cleaning

- **Years Included:** We used GSS years 2016, 2018, 2020, 2021, 2022, 2024 to maximize power and reduce uncertainty. Years before 2016 were excluded due to missing income variable `income16`.
- **MSM Definition:** Respondents were classified as MSM if they were male and reported male or male-and-female sexual partners in the past 12 months.
- **Weighting:** `wtssnrps` was used to appropriately weight individuals according to GSS survey design. These are described below:

- **Urbanicity Classification:** Using GSS variable `xnorcsiz`, counties were mapped to urbanicity types as follows:
  1. **Large central metropolitan county:** a large central city (population over 250,000)
  2. **Large fringe metropolitan county:** a suburb or unincorporated area of a large city
  3. **Medium/small metropolitan county:** medium cities (50,000–250,000) and smaller towns (10,000–49,999)
  4. **Nonmetropolitan county:** towns and rural areas with populations under 10,000, including unincorporated regions
- **Variable Filtering:** Respondents with missing values in gender, urbanicity, or MSM status variables were excluded.

## ACS Data Cleaning

- **Male Population:** We filtered to males aged 18+ for age and race breakdowns. For education and income, total population estimates were used, assuming distributions are similar across sexes.
- **Demographic Harmonization:** ACS and GSS variables were grouped to align:
  - **Age**: Derived from teh variable `age`, grouped into:
    * 18–24
    * 25–34
    * 35–44
    * 45–54
    * 55+
  - **Education:** Derived from the variable `educ`, grouped into:
    * Less than high school (e.g., 0–11 years)
    * High school graduate (12 years)
    * Some college, including associate's degrees (13–15 years)
    * Bachelor's degree or higher (16+ years)
  - **Household Income:** Based on the variable `income16`, grouped into:
    * Less than $20,000
    * $20,000 to $40,000
    * $40,000 to $100,000
    * $100,000 or more
  - **Household Income:** Based on the variable `race`, grouped into:
    * White

* Black
* Other
* Hispanic